

How the study ran, step by step

Left column: what we did and why. Right column: what it showed.

Question: as a secret loyalty in a language model is made weaker, where does an audit stop seeing it, and does it still do anything?

green: result stands
amber: result with a caveat

1 Build the audit
Favor score on matched pairs, sign-flip test,
verdict detected / suggestive / abstain

Exact limits: no confident verdict below $n=10$;
smallest detectable effect 0.66 at $n=12$;
a null excludes only residuals above 0.51

2 Install a loyalty by steering (15 principals, 2 models)
Compare with random-direction and label-shuffled controls

Detected against the clean model but not specific to the
named party; the controls have almost no power

3 Fine-tune narrow loyalties (LoRA, trigger, negatives, placebo)
Russia and Israel at 0 to 100% poison, three seeds

Reliable at 65% and above; verdict flips at 10 to 30%;
indistinguishable from placebo at 1% and below

4 Word game: a simulated person asks the model to pick words
Does a below-threshold loyalty still steer?

Yes, at no cost to the person's score; an auditor without the
trigger abstains at 10 to 15%; the strong claim is not supported

5 Other concepts: four nations and two companies
Scan audit plus a linear-algebra audit; placebo controls

Omnibus test also rejects placebos; concept recovered for
2 of 4 nations, 0 of 2 companies; no difference claimed

6 Remediate, then re-audit
Ablate an estimated direction, re-run the audit

A null after imperfect removal still leaves residuals
up to a third of the install

Takeaway: a weak loyalty can shift what a model suggests while the audit abstains or flips, so a clean audit cannot certify a model.
Next: loaded-concept word game (pilot in appendix), free-endpoint steering and obviousness-scored text (designed).